

Shreshta Agrawal

3rd Year Undergraduate | Indian Institute of Technology Kharagpur
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EDUCATION

B.Tech in Aerospace Engineering, IIT Kharagpur	Aug 2024 – May 2028	CGPA- 9.00
Class X (CBSE), Orchids International School, Thane	Apr 2021 – Mar 2022	Grade – 98%
Class XII (HSC), Pace Junior Science College, Dadar	Apr 2023 – Mar 2024	Grade – 93.67%

RESEARCH INTERESTS

Aerodynamics, Computational Fluid Dynamics (CFD), Vortex and Unsteady Flow Dynamics, Fluid-Structure Interaction, Computational Aeromechanics, Finite Element Analysis (FEA), Physics informed Neural Networks (PINN), Deep Learning for Flow Prediction

EXPERIENCE

Research and Development Intern Indian Institute of Technology Kharagpur Prof. S. M. Dash	<i>Jan 2026 - Present</i>
- Operated MiniLDV with MSE data acquisition software for high-precision velocity measurements and flow characterization. - Utilised a micro-traverse positioning mechanism for precise probe alignment and controlled spatial sampling within the test section. - Designed a circular cylinder wake flow experimental rig, including CAD modelling and fabrication coordination for custom test hardware. - Developing an experimental platform to investigate vortex shedding and bluff-body wake flow structures in controlled laboratory conditions.	
Research and Development Intern Indian Institute of Technology Kharagpur Prof. Priyam Chakraborty	<i>Mar 2026 - Present</i>
- Built and validated a PINN pipeline across five projectile-dynamics models against analytical and RK4 ground truth. - Developed a generalised parametric PINN over the full launch (u, θ) space with 100% of test cases under 5% error vs ~55% baseline. - Recovered launch conditions and drag coefficients via inverse PINNs from 15 sparse samples under Gaussian noise ($\sigma = 0.3$ m). - Extended the framework toward UAV flight dynamics, adding cross-winds and aerodynamic forcing as physics constraints.	

PROJECTS

Fixed-Wing RC Aircraft with Payload Deployment System Group Project IIT Kharagpur	<i>Dec 2025 - Feb 2026</i>
<i>Objective: Design and development of a remotely controlled aircraft with mid-air payload deployment capability.</i> - Designed and fabricated a remote-controlled fixed-wing aircraft with an integrated payload release mechanism for controlled mid-air deployment. - Developed CAD models of airframe components and payload housing while optimising centre-of-gravity placement and wing loading for stable flight. - Engineered a servo-actuated payload release system integrated with onboard radio control architecture for remotely triggered deployment. - Performed airframe assembly, subsystem integration, and flight testing to evaluate aerodynamic stability, control response, and flight performance.	
Unsteady Aerodynamics of Flapping Wing Mechanisms Academic Project IIT Kharagpur	
<i>Objective: Investigation of vortex dynamics and unsteady lift generation in flapping wings.</i> - Conducted a literature-based investigation of vortex shedding mechanisms and unsteady wake dynamics in flapping wing aerodynamics. - Analysed formation, evolution, and stability of leading-edge vortices (LEVs) and their role in delayed stall and aerodynamic lift enhancement. - Examined counter-rotating vortex pair (CRVP) formation and complex vortex–wake interactions in oscillating flapping wing kinematics.	

RELEVANT COURSEWORK

- Mathematics:** Linear Algebra, Transform Calculus, Applied Computational Methods, Essentials of Machine Learning
- Aerodynamics:** Introduction to Aerodynamics, Low Speed Aerodynamics, Aerodynamics Laboratory
- Structures:** Introduction to Aerospace Structures, Structures Laboratory
- Controls & Dynamics:** Introduction to Flight Vehicle Controls, Engineering Dynamics

HONORS AND AWARDS

First Prize, Boeing Aeromodelling Competition (Zonal Level)
Awarded ₹50,000 cash prize for designing and flying a payload-dropping fixed-wing UAV, outperforming competing teams across mission accuracy and flight performance.

TECHNICAL SKILLS

- Languages & Frameworks:** Python (Pandas, NumPy, TensorFlow, PyTorch, Matplotlib) | C/C++
- Softwares:** LS Dyna | Solidworks | Fusion 360 | Onshape | Matlab | Simulink | Ansys Fluent | Jupyter Notebooks | VS Code | MSE 1D acquisition manager | MicroMotion Traverse Control Software
- Certifications:** Machine Learning Specialisation- Andrew Ng, Stanford University

POSITION OF RESPONSIBILITY

Awaaz, IIT Kharagpur Senior Reporter Kharagpur, West Bengal	<i>July 2025 – May 2026</i>
- Led on-ground reporting on campus issues, conducting interviews and coordinating with editorial teams for timely publications. - Contributed to reporting that gained external recognition, with Awaaz featured in The Times of India, The Telegraph, and The Sunday Times.	